



AIML

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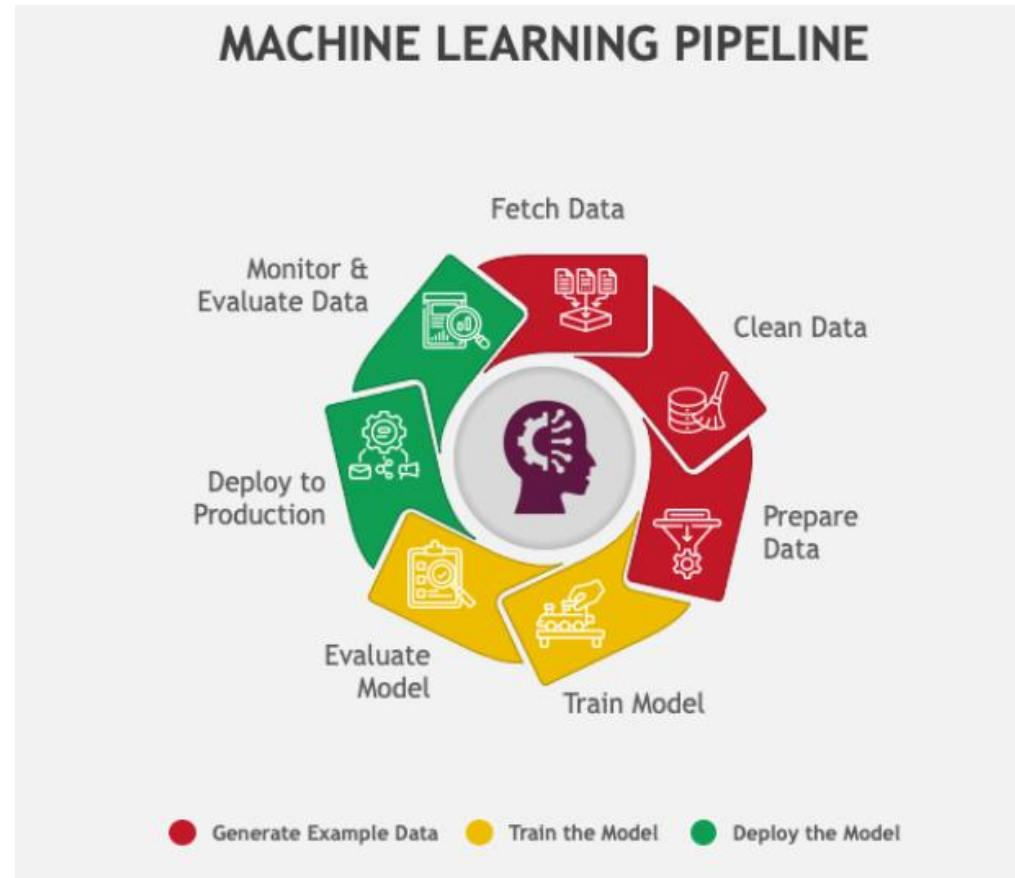
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AI Landscape

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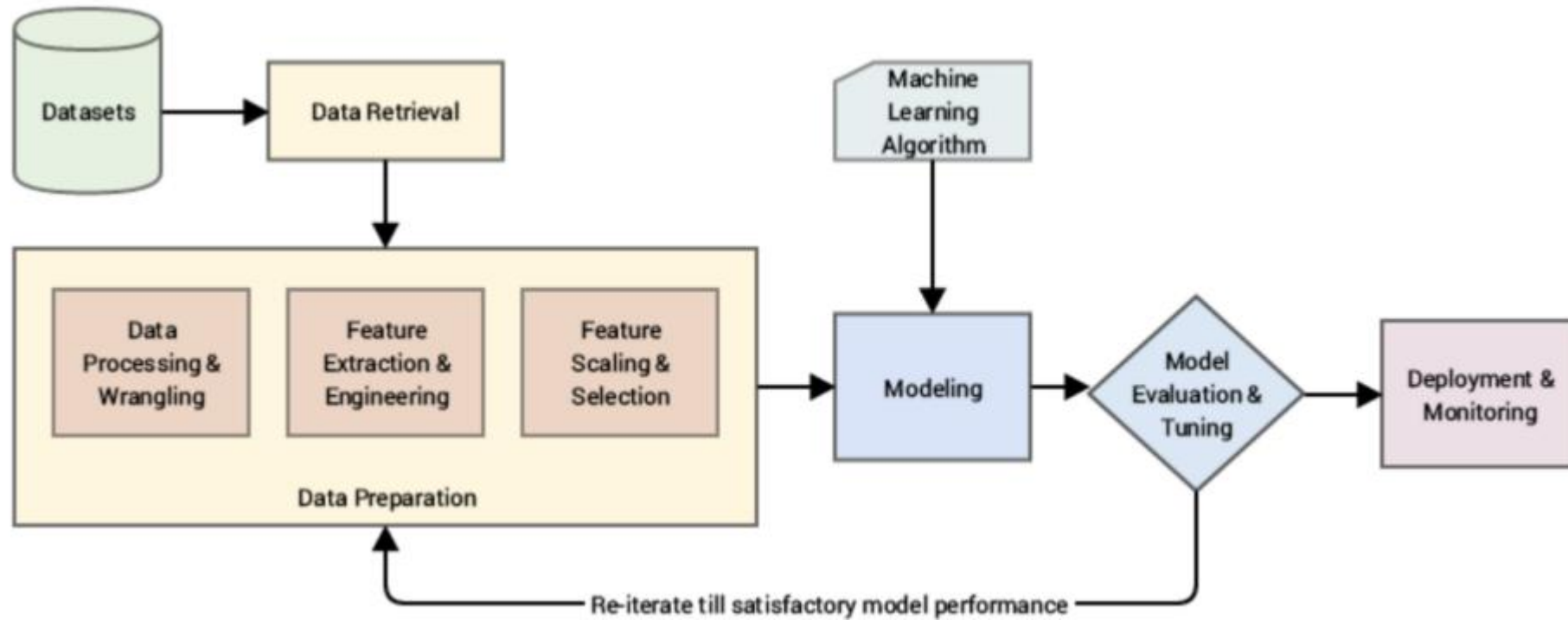
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A **Machine Learning (ML) Pipeline** is a structured workflow that guides the stages involved in creating a machine learning model.



1. Fetch Data: Collect the data relevant to problem statement
2. Data Cleaning: Impute missing values, Remove duplicate records, Remove Outliers
3. Data Preparation: Process and prepare the data using techniques such as:
 - Feature Scaling: Normalize values (Bring numeric values to a common scale)
 - Label Encoding: change the categorical value into numerical format
 - Feature Selection: Select the significant/correlated features
 - Feature extraction: Transforming the data into simplified features
 - Data Splitting: Split the data into a Train data and Test data

4. Train Model: Train the machine models against the training dataset using appropriate machine learning algorithms
5. Evaluate Model: Measure the performance of the models using test and validation subsets of data to determine model accuracy
6. Model Deployment: Make the trained model available for real-world which can be used for Prediction
7. Performance Monitoring: Continuously monitor model performance, retrain, and calibrate accordingly. Collect new data to continue to improve the model and prevent it from becoming stale





THANK YOU

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